

S-Move Dealer — Scaling Limits & Cost Plan

Prepared 2026-06-30. Pricing verified against current Supabase & Railway published rates.

TL;DR

- **Users are NOT a cost.** Dealer uses custom token auth, zero Supabase Auth — so MAU billing (the metric tied to "thousands of users") is £0. Drop that worry.
- **The real cost driver is image EGRESS (bandwidth), not storage volume.**
- **Images are stored & served at full resolution with no compression or thumbnails.** This is the single highest-leverage fix — it cuts storage AND egress 5–10×.
- With an image pipeline in place, infra runs ~£60–120/mo early and ~£150–350/mo at 500 businesses. Without it, egress can triple that.
- Claude API + Apify are variable spend you *control* (throttles), not structural ceilings.

1. Current architecture (facts)

- **DB:** Supabase Postgres (`wayqlbhdflyioacqbwbtb`), 45+ tables. Rows are small (items ~300–500 B, images ~150 B).
- **Images:** Supabase Storage buckets `item-images`, `item-drafts`, `seller-logos`. Multer caps: 20 MB/file, **10 images/item. No resize, no compression, no thumbnails** — full-res stored and served.
- **Hosting:** Single Railway Node/Express service (`shop.s-move.co.uk`, port 3001).
- **Background work (all in-process, no job queue):**
 - Engagement scoring every 30 min — UPDATES every active item.
 - Continuous eBay scraper loop.
 - On-demand Apify scrapes (FB Marketplace, eBay sold).
 - Claude calls on listing analysis, moderation, drifting-item tips, chat.
- **Auth:** custom token tables for admin / seller / customer. **No Supabase Auth → no MAU charge.**

2. Limits, in the order they bite

Limit	Included (Supabase Pro \$25/mo)	Bites when	Overage cost
Image egress	250 GB/mo unified	A few hundred users browsing full-res catalogue	\$0.09/GB uncached, \$0.03/GB cached
Railway single instance	1 process	Scrapers + scoring loop saturate CPU/memory	reliability wall first
File storage volume	100 GB	~30k listings at full res	~\$0.021/GB/mo (cheap)

Limit	Included (Supabase Pro \$25/mo)	Bites when	Overage cost
Claude + Apify spend	usage only	scales with listing velocity & scan frequency	variable, controllable
DB storage	8 GB	years out (tiny rows)	\$0.125/GB/mo
Supabase MAU	100k	never (unused)	—

3. Cost at three growth stages

Assumes ~100 products/business + individual sellers, ~4 images each. **As-is vs. with image pipeline.**

Monthly	A: ~50 biz (~6k listings)	B: ~200 biz (~25k)	C: ~500 biz (~65k)
Supabase base	£20	£20	£20
Storage as-is	within	+£5	+£15
Storage compressed	within	within	within
Egress as-is	£30–80	£150–400	£400–1,000+
Egress compressed	within	£20–60	£80–250
Railway compute	£10–20	£30–50	£50–100
Claude API	£30–150	£80–400	£150–800
Apify	£120–400 (throttle)	same	same
Stripe	2.9% + 30p passthrough, scales w/ revenue		

Infra-only (excl. Claude/Apify), pipeline fixed: ~£60–120/mo → ~£150–350/mo at scale.

Image storage maths (full-res, ~3 MB avg, 4/listing): - A: ~24k images ≈ 72 GB · B: ~100k ≈ 300 GB · C: ~260k ≈ 780 GB - Compressed + thumbnails (~600 KB stored): A ≈ 15 GB · B ≈ 60 GB · C ≈ 150 GB

4. Action list (priority order)

- 1. Image pipeline — DO THIS BEFORE LAUNCH SCALE.** On upload: resize to a sane max (e.g. 1600px long edge), compress to WebP/JPEG ~80%, generate a thumbnail (~400px) for grid/cards. Serve thumbnails in listings, full image only on detail view. Highest ROI by far — cuts egress + storage 5–10×.
- 2. Put Supabase Storage behind a CDN / use cached egress.** Cached egress is \$0.03/GB vs \$0.09 uncached — 3× cheaper. Set long cache headers on `item-images` (immutable filenames already help).
- 3. Retention policy on unbounded tables.** `item_views`, `search_terms`, `external_sold_prices`, `market_comparisons` grow forever. Archive/trim `item_views` > 90 days and `search_terms` monthly. Not urgent for cost (rows are small) but keeps queries fast.
- 4. Move background work off the web instance.** The 30-min scoring loop + continuous eBay scraper + Apify runs share one process with live traffic. At scale, split into a separate

Railway worker service (or cron) so a scrape can't slow the shop or a restart can't stall scoring.

5. **Throttle / budget the variable APIs.** Cap drifting-item tip generation (worst-case spike), and gate Apify scans behind explicit triggers with a monthly credit ceiling. These are the spend wildcards.
 6. **Watch egress weekly once live.** It's the metric that turns a £25 bill into a £250 bill silently. Set a Supabase usage alert.
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5. What you do NOT need to worry about

- Number of registered users (no MAU billing).
- Database storage volume (tiny rows, years of runway on 8 GB).
- Number of listings as a hard cap (Postgres handles 65k items trivially; it's the *images attached* that cost).